

From Timbre to Symbol: Semiotic Feedback Loops in Machine-Augmented Improvisation

Artistic Report

1. Introduction: Rethinking Musical Signs Beyond the Score

Western art music has historically privileged the score as the primary site of musical knowledge. Notation functions as a symbolic system that precedes sound, organising pitch, rhythm, harmony, and form before any performance takes place. Even in jazz and improvised traditions, where the score may be reduced, flexible, or absent altogether, symbolic abstractions such as chord symbols, lead sheets, and formal schemas often continue to underpin planned musical actions.

Yet much of what musicians actually respond to in real-time performance, particularly in free improvisation and much contemporary jazz practice, does not reside in symbolic notation. Instead, it emerges from the immediate sonic environment: timbre, density, articulation, gesture, and energetic change.

This artistic report will demonstrate the online work of the Deep Improv project that explores these sonic dimensions as primary carriers of musical meaning through a machine-augmented improvisation system. Rather than analysing pitch or harmony directly, the system listens to the timbral behaviour of performers, including changes to sound parameters such as those produced through instrumental technique or live signal processing (for example, adjustments to effects pedals). These continuous timbral changes are analysed using machine learning models trained on data drawn from current and past performances. The system translates this information into audio- and MIDI-based cues that are reintroduced into the performance as prompts for further musical action. In addition to these timbre-driven cues, the system may introduce idiomatic references drawn from jazz practice, such as common harmonic progressions or short audio samples derived from well-known performances. These cues are not intended to prescribe musical outcomes, but to function as interpretive prompts within the improvisational flow. The work will be presented as a live lecture-demonstration, combining explanation, performance, and reflection, and includes live improvised performance.

2. Semiotics, Improvisation, and the Neutral Level of Sound

The central artistic and theoretical proposition of this work is that musical signs need not originate in the score. Instead, they can emerge dynamically from sound itself, circulate through machine listening systems, and return to performers as actionable symbols. Drawing on Jean-Jacques Nattiez's tripartite semiotic model of musical meaning (1990), Deep Improv can be understood as a deliberately constructed semiotic feedback loop, in which sonic traces are analysed at the neutral level, then re-encoded symbolically, and then reinterpreted at the esthetic and poietic levels during performance (see Fig. 1).

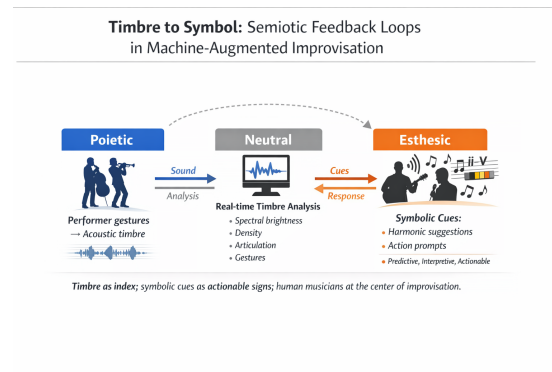


Figure 1. Semiotic feedback loop in Deep Improv. Continuous timbral features at the neutral level of the sonic trace are analysed by a machine learning system and translated into audio and MIDI cues. These cues are interpreted by human performers at the esthetic and poietic levels, influencing subsequent musical action and generating new sonic material.

Nattiez’s theory of musical semiotics distinguishes between three analytically separable levels: the poietic (the processes of creation), the neutral (the material trace, such as the sounding music or written score), and the esthetic (the processes of perception and interpretation). While music analysis has often focused on symbolic structures at the poietic level, such as compositional intent, formal organisation, harmonic grammar, Nattiez emphasises that meaning emerges through the interaction of all three levels and cannot be reduced to any single domain.

In improvised music, this interaction is especially fluid. The poietic and esthetic processes of performers unfold simultaneously, and the neutral level is no longer a static score to be followed, but a continuously evolving sonic field. Performers respond not only to what has already occurred, but to what the sound seems to indicate or foreshadow. These anticipatory interpretations, which might be described informally as “where the music wants to go” is rarely encoded explicitly, yet they play a crucial role in shaping improvisational decision-making. Research in embodied music cognition and ecological approaches to listening further supports the view that musicians extract meaningful structure directly from sonic interaction rather than symbolic abstraction alone (Leman, 2007; Windsor, 2012).

Deep Improv intervenes precisely at this point. By treating the sounding music itself as a field of semiotic cues, the system makes explicit what is normally tacit: the inferential processes through which experienced improvisers read timbral information as signs of tension, release, continuation, or transformation. In doing so, it reframes improvisation not as the absence of structure, but as a dense semiotic activity grounded in real-time listening.

3. System Overview: Timbre as Input, Symbol as Output

The Deep Improv system operates as a real-time, machine-augmented listening environment that processes both live audio and MIDI data from one or more performers. Incoming audio streams are analysed continuously to extract features related to timbre, gesture, and texture, including spectral centroid (perceived brightness), spectral flux, dynamic density, articulation rates, and measures of textural thickness. Where available, incoming MIDI data—such as note density, velocity distribution, controller activity, and temporal clustering—is analysed in parallel and used to contextualise and stabilise audio-derived inferences.

These feature streams are integrated within a machine learning framework trained on data collected from prior improvisations, allowing the system to model correlations between evolving timbral behaviour and higher-level musical tendencies. Feature-based approaches of this kind are well established in research on timbre perception, music information retrieval, and electroacoustic analysis (Grey, 1977; Peeters, 2004), and are here adapted specifically for real-time, performative interaction rather than post hoc analysis.

Crucially, the system does not attempt to infer explicit pitches, chords, or formal sections. Instead, it prioritises how the music is unfolding, such as its energetic profile, gestural direction, and textural evolution, rather than what notes are being played at any given moment. This design choice reflects the project's grounding in free and jazz-informed improvisation practices, in which timbral change and gestural development often precede, imply, or even override harmonic specification as primary drivers of musical meaning (Smalley, 1997; Wishart, 1996).

The analysed feature data are mapped to a set of symbolic outputs that function as performance cues. These outputs take several forms:

- MIDI-based cues that suggest changes to parameters being used on effects pedals
- Audio cues, including short idiomatic fragments or gestural references derived from jazz vocabulary and historically situated recordings
- Textual or graphical indicators that represent emergent tendencies toward continuation, rupture, accumulation, or stabilisation within the improvisation

These outputs are intentionally underspecified and non-prescriptive. They do not function as compositional instructions or deterministic controls. Instead, they operate as high-level semiotic triggers that rely on shared cultural knowledge and performer expertise to become meaningful in practice. By translating continuous timbral and gestural data into symbolic prompts, the system externalises interpretive listening processes and reintroduces them into the improvisational field as material for creative response.

4. Timbre as Index, Harmony as Symbol

From a semiotic perspective, the distinction between timbral input and harmonic output is crucial. Following the Charles Peirce terminology as adopted by Nattiez for use in music, timbral features function primarily as indices: they point to physical actions, energy states, and embodied gestures. A sudden increase in spectral brightness may index heightened intensity or instrumental strain; dense textures may index collective momentum or saturation. Similar indexical interpretations of sound have been explored in gesture-based and spectromorphological accounts of listening (Godøy, 2010; Smalley, 1997).

The harmonic cues produced by the system, by contrast, function symbolically. They are culturally learned signs whose meaning depends on shared conventions within jazz and related improvisational traditions. A ii–V implication does not describe a specific sound, but rather invokes a network of expectations concerning tension, resolution, phrasing, and formal direction (Hatten, 2004).

In Deep Improv, indexical timbral signs are translated into symbolic harmonic cues, creating a cross-domain semiotic transformation (Fig. 1). This translation does not claim objectivity or correctness. Rather, it deliberately foregrounds the interpretive leap involved in musical listening. The idea is that the machine does not “know” where the music is going; it infers tendencies based

on learned correlations, mirroring the way human improvisers anticipate musical trajectories before they are realised.

5. Predictive Listening and Anticipation in Jazz Practice

The system's emphasis on prediction rather than reaction aligns closely with established accounts of jazz improvisation. Ethnographic studies by Berliner (1994) and Monson (1996) describe how experienced musicians continually project future possibilities while playing, responding not only to present sounds but to implied directions and emergent forms. More recent work in improvisation studies and computational creativity similarly emphasises anticipation as a core component of real-time musical interaction (Sawyer, 2006; Lewis, 2000).

Moments where the system's inference diverges from human intention are treated as productive. Misalignment can be used as a site of creative tension, revealing the contingency of musical meaning and the multiplicity of possible interpretations that always coexist within any improvised moment.

6. From Visual Notation to Sonic Cues for Performers

A key contribution of the work lies in its rethinking of musical notation. Traditional notation externalises musical structure visually, prior to sound. Even graphic and experimental scores, while expanding symbolic vocabulary, retain a visual bias. In contrast, Deep Improv proposes a form of sonic notation, in which cues emerge from sound itself and are returned to performers within the temporal flow of performance. This idea resonates with broader critiques of notation-centred epistemologies in music theory and analysis (Cook, 2013; Ingarden, 1986).

This approach aligns with historical cue-based and game-based improvisation systems (e.g. Butch Morris's Conduction, Zorn's Cobra), yet differs in that the cues are generated through machine listening rather than human conduction. The machine functions as a semiotic mediator, translating between sonic indices and symbolic prompts without occupying a traditional authorial role.

7. Performance Practice and Artistic Experience

The live demonstration includes up to ten minutes of improvisation, during which performers hear and respond to the system's cues in real time. The experience of playing with Deep Improv is intentionally ambiguous. Performers report heightened attention to timbral detail, increased awareness of collective texture, and a shifting sense of agency as machine-generated prompts enter the improvisational dialogue. Comparable observations have been made in studies of interactive music systems that foreground listening over control (Bown et al., 2015).

Crucially, performers are not required to follow the cues. Ignoring or contradicting them is an equally valid response. This openness preserves the ethical and aesthetic centrality of the human performer, positioning the machine as a participant rather than an authority.

The system thus acts less as a generative composer and more as an interpretive mirror, reflecting aspects of the music back to the ensemble in altered symbolic form. The artistic value of the work emerges not from technical novelty alone, but from how it reshapes listening, attention, and interaction within improvised performance.

8. Contributions and Broader Implications

Deep Improv contributes to ongoing discussions in timbre studies, improvisation research, human–computer interaction, and the development of symbolic and graphical representations of sound. By grounding its operation in semiotic theory, the project offers a conceptual framework for understanding machine listening not merely as signal processing, but as an interpretive act embedded within cultural practice.

More broadly, the work suggests a pathway for re-centering human musicianship in machine-augmented performance. Rather than aiming for stylistic imitation or autonomous generation, the system treats musical knowledge as a shared semiotic resource, activated through listening and interpretation.

The author proposes the work for consideration in an acoustic or electroacoustic concert setting, subject to selection by the committee and technical feasibility. Any required performers will be provided by the author.

9. References

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